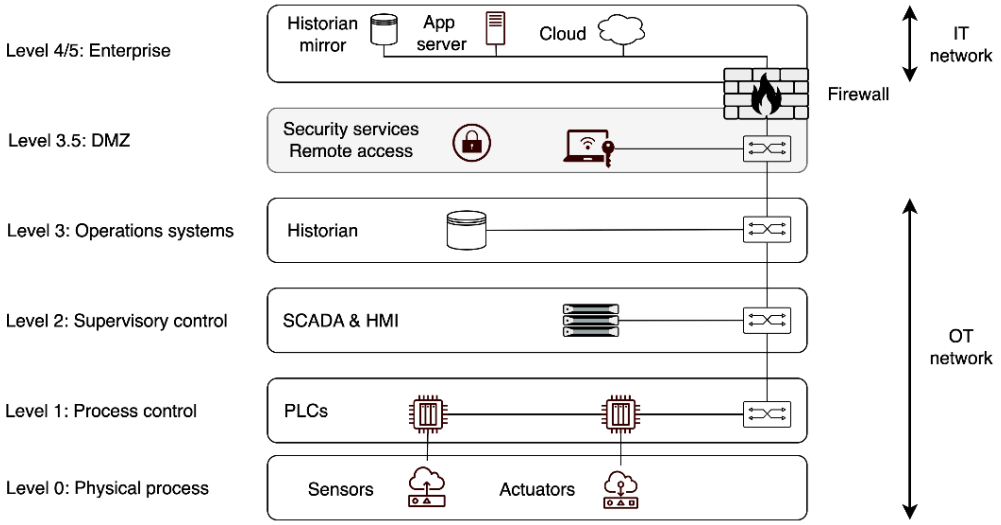


Figure 4.5 The Purdue model of IT/OT convergence segments the network into layers with a DMZ acting as a buffer zone between the OT and IT networks.

4.2.4 External data sources

Digital twins frequently transcend the confines of a single organization, relying heavily on data that originates from external sources beyond your direct control. These external data streams can range from third-party APIs and public datasets to partner organizations' systems and cloud-based services. In many cases, external systems serve as the primary, or even exclusive, source of important information for your digital twin implementation. More commonly, external data supplements and enriches your internal datasets, providing valuable contextual information that enhances the accuracy and comprehensiveness of your digital twin models.

In section 4.4 we will look at different ways that external data can be integrated into your digital twin.

4.3 Data structures

How you structure data for storage and processing significantly impacts query performance, storage efficiency, and analytical capabilities. The architectural decisions you make about data organization—whether to prioritize row-based or columnar storage, how to partition datasets, what indexing strategies to employ, and how to handle schema evolution—will determine not only how quickly you can retrieve information but also what types of analysis become feasible at scale. Different structural approaches optimize for different access patterns and use cases, creating trade-offs between write performance and read efficiency, storage compression and query flexibility, or real-time processing and historical analysis.